

Suppose a man who is drunk is trying to cross a bridge 5 feet wide with no railing on either side. Let X_n be the distance measured from the left edge of the bridge in feet in the n^{th} state. The drunkard can move either 1 foot left or same distance right with probability p and $q = 1 - p$ respectively. However he could not move forward or backward although he intends to cross the bridge say 10 feet long. When he reaches the edge of the bridge either to the left ($X_n = 0$) or to the right ($X_n = 5$) he falls in the river and our interest in observing his random walk ends. Describe the process diagrammatically and hence establish the transition matrix for the random walk problem.

1.

Let X_n be the whether conditions on the n^{th} day at Mumbai. Suppose there are three states say 1–Pleasant, 2–Bad and 3–Fair. Subject to the availability of following transition matrix

$$P = \begin{bmatrix} 0.5 & 0.4 & 0.1 \\ 0.4 & 0.5 & 0.1 \\ 0.3 & 0.5 & 0.3 \end{bmatrix}$$

Then Answer the questions given below:

- 1 Suppose today is Monday and it is bad, what is the probability that Tuesday is fair and Wednesday is pleasant?
- 2 What is the probability that Thrusday is pleasant given that Tuesday is bad?

2.

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The transition probability matrix of a Markov Chain $\{X_n\}_n \in N$

with three states 0,1,2 is as follows: $P = \begin{bmatrix} \frac{3}{4} & \frac{1}{4} & 0 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{2}{3} & \frac{1}{4} \\ 0 & \frac{1}{4} & \frac{1}{4} \end{bmatrix}$, the initial

distribution is $P^{(0)} = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}$ then find

$$P(X_2 = 2), P(X_3 = 1, X_2 = 2, X_1 = 1, X_0 = 2)$$

Ans: $\left[\frac{1}{6}, \frac{3}{64} \right]$

3.

4. (Forecasting the Weather): Suppose that the chance of rain tomorrow depends on previous weather conditions only through whether or not it is raining today and not on past weather conditions. Suppose also that if it rains today, then it will rain tomorrow with probability a ; and if it does not rain today, then it will rain tomorrow with probability b . If we say that the process is in state 0 when it rains and state 1 when it does not rain, then the above is a two-state Markov chain whose transition probabilities are given by

$$P = \begin{bmatrix} a & 1-a \\ b & 1-b \end{bmatrix}$$

Considering the above situation in which the weather is considered as a two-state Markov chain. If $a = 0.7$ and $b = 0.4$, then calculate the probability that it will rain four days from today given that it is raining today.

5. (Transforming a Process into a Markov Chain): Suppose that whether or not it rains today depends on previous weather conditions through the last two days. Specifically, suppose that if it has rained for the past two days, then it will rain tomorrow with probability 0.7; if it rained today but not yesterday, then it will rain tomorrow with probability 0.5; if it rained yesterday but not today, then it will rain tomorrow with probability 0.4; if it has not rained in the past two days, then it will rain tomorrow with probability 0.2. If we let the state at time n depend only on whether or not it is raining at time n , then the above model is not a Markov chain (why not?). However, we can transform the above model into a Markov chain by saying that the state at any time is determined by the weather conditions during both that day and the previous day. In other words, we can say that the process is in state 0 if it rained both today and yesterday, state 1 if it rained today but not yesterday, state 2 if it rained yesterday but not today, state 3 if it did not rain either yesterday or today.

Consider above situation. Given that it rained on Monday and Tuesday, what is the probability that it will rain on Thursday?